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Conditional Cash Transfers as Stabilization Tool: a HANK Approach

1. Model

The setup is a standard New Keynesian model with incomplete markets and labor supply risk. Time is discrete and runs from $t = 0$ to infinity. There are four types of agents in the economy: households, firms, labor unions, and the government.

Households are heterogeneous (Aiyagari 1994), workers and firms join in a formal-informal setup, and the rest of the economy follows the textbook New Keynesian model (Galí 2015). There is no aggregate uncertainty and the economy starts in the deterministic steady-state.

Households

Assume there exists a continuum $i \in [0, 1]$ of ex-ante identical households, but experiencing idiosyncratic shocks in labor productivity and discount factor.

Households enjoy a separable CRRA utility over consumption $c_{i,t}$ and worked hours $h_{i,t}$, which we define $u(c_{i,t}, v_{i,t}) = u(c_{i,t}) - v(h_{i,t})$ as

$$\max_{c_{i,t},h_{i,t}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_{i,t}^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} - \psi \frac{h_{i,t}^{1+\frac{1}{\varphi}}}{1+\frac{1}{\varphi}} \right],$$

where γ is the elasticity of intertemporal substitution (EIS) and φ is the Frisch elasticity of labor supply.

A household faces idiosyncratic labor productivity shocks $e_{i,t}$ with $\mathbb{E}[e_{i,t}] = 1$, and discount factor heterogeneity $\beta_i \in \{\beta_L, \beta_H\}$, and we fix ω_I impatient households. They can work in the formal or informal sector, or be unemployed $s \in \{F, I, U\}$. Each sector has a specific productivity $\theta^s_{i,t}$ which their wage depends on, re-drawn on each new offer, with $\mathbb{E}\left[\theta^F_{i,t}\right] = 1$.

Formal workers earn labor income $(1 - \tau_\ell) w_t h_{i,t} e_{i,t} \theta^F_{i,t}$, while informal workers earn $w^I_t h^I_{i,t} e_{i,t} \theta^I_{i,t}$, where w_t and w^I_t are the baseline wage, τ_ℓ is a fixed labor income tax rate.

Unemployed earn a subsistence income y_t^U. Let $y_t(s, e, \theta)$ be this non-financial income,

$$y_t(s,e,\theta) = \begin{cases} (1-\tau_\ell) w_t h_t e \theta^F, & \text{if formal;} \\ w_t^I h_t^I e \theta^I, & \text{if informal;} \\ y_t^U, & \text{if unemployed.} \end{cases}$$

Let $b_{i,t} \in \{0,1\}$ be the *Bolsa Família* (BF) status, so that the transfer size is $T_{i,t} = T_t b_{i,t}$. At the end of each period, non-recipients in informality or unemployment enter with probability λ if eligible by an income criterium $y_{i,t} + r_t^a a_{i,t} < \bar{y}$, where y^U does not count for means test. At the same time, beneficiaries face a probability ρ_s of being audited, where $\rho_F > \rho_I$.

We assume that audited formal recipients always leave, while informal and unemployed households stay only if eligible. Formal income is cross-checked automatically, however, the lag between entering a formal job and losing the benefit motivates a $\rho_F < 1$. Therefore, the probability of being a recipient next period is

$$\mathbb{P}\left[b' = 1 | s, b, \theta, e, a\right] = q(s,b,\theta,e,a) = \begin{cases} \lambda \cdot \mathbb{1}\{s \neq F, y + ra < \bar{y}\}, & \text{if } b = 0 \\ (1-\rho_s) + \rho_s \mathbb{1}\{s \neq F, y + ra < \bar{y}\}, & \text{if } b = 1 \end{cases}.$$

However, we assume that assets are hidden from the government, so the agency approves within a group of observationally identical households, given by $x = (s, b, \theta, \beta, e)$. Thus, the agency approves with a conditional probability $\bar{Q}(x, b)$, weighted by the assets distribution, given by

$$\bar{Q}(x) = \mathbb{E}\left[q(x,a)|x\right].$$

Households save into a mutual fund of liquid assets $a_{i,t}$ with ex-post real return $1 + r_t^a$, and pays consumption taxes τ_t^c. Their budget constraint at time t reads:

$$(1-\tau_t^c) c_{i,t} + a_{i,t} = (1 + r_t^a) a_{i,t-1} + y_{i,t}(s,e,\theta) + T_t b_{i,t}.$$

Incomplete markets impose a borrowing constraint $a_{i,t} \geq \underline{a}$. Frictions in the labor market imply that workers delegate their labor supply decisions to labor unions, and thus take worked hours as given, so their labor disutility is internalized at the union level, not by individual households, such as Motta (2025).

Such as Esteban-Pretel and Kitao (2021) and Finamor (2025), given a sector s, we assume that each worker faces an exogenous layoff probability δ_s, and receives an offer from s with probability π_s if employed or π_s^U if unemployed. The household accepts the offer with the higher value, smoothed by additive idiosyncratic preference shocks explained in the Appendix.

The Bellman Equation for each individual i in sector $s \in \{F, I\}$ is

$$
\begin{aligned}
V_t^s(a, e, \beta, \theta) = \max_{c, a'} \Bigg\{ & \frac{c^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} - \psi \frac{h^{1+\frac{1}{\varphi}}}{1+\frac{1}{\varphi}} + \beta \mathbb{E}_t \Big[\delta_s V_{t+1}^U(.) + (1-\delta_s) \Big(\\
& \pi_F (1-\pi_I) \max\Big\{ V_{t+1}^s(., \theta), V_{t+1}^F(., \theta') \Big\} \\
& + (1-\pi_F) \pi_I \max\Big\{ V_{t+1}^s(., \theta), V_{t+1}^I(., \theta') \Big\} \\
& + \pi_F \pi_I \max\Big\{ V_{t+1}^s(., \theta), V_{t+1}^I(., \theta'), V_{t+1}^F(., \theta') \Big\} \\
& + (1-\pi_F)(1-\pi_I) V_{t+1}^s(., \theta) \Big) \Big] \Bigg\} \\
\text{s.t.} \quad & (1-\tau_t^c) c + a' = (1+r_t^a) a + y_t(s, e, \theta) + T_t b \\
& a' \in [\underline{a}, \infty),
\end{aligned}
$$

where workers take hours as given. And for each unemployed worker is

$$
\begin{aligned}
V_t^U(a, e, \beta) = \max_{c, a'} \Bigg\{ & \frac{c^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} + \beta \mathbb{E}_t \Big[(1-\pi_F^U)(1-\pi_I^U) V_{t+1}^U(.) \\
& + \pi_F^U (1-\pi_I^U) \max\Big\{ V_{t+1}^U(.), V_{t+1}^F(.) \Big\} \\
& + (1-\pi_F^U) \pi_I^U \max\Big\{ V_{t+1}^U(.), V_{t+1}^I(.) \Big\} \\
& + \pi_F^U \pi_I^U \max\Big\{ V_{t+1}^U(.), V_{t+1}^I(.), V_{t+1}^F(.) \Big\} \Big] \Bigg\} \\
\text{s.t.} \quad & (1-\tau_t^c) c + a' = (1+r_t^a) a + y_t(U) + T_t b \\
& a' \in [\underline{a}, \infty).
\end{aligned}
$$

We denominate the steady-state size of each sector $s \in \{F, I, U\}$ as α_s, such that $\alpha_F + \alpha_I + \alpha_U = 1$. Let $\Lambda_t(s, b, \theta, \beta, e, a)$ be the household distribution over a given period of time.

Firms

In the firm side, there is a representative final-good producer that aggregates a continuum of $j \in [0, 1]$ intermediate inputs produced by formal firms,

$$
Y_t = \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} \, dj \right)^{\frac{\varepsilon}{\varepsilon-1}},
$$

where $\varepsilon > 0$ is a constant substitution elasticity (CES).

Each intermediate good j is produced by a monopolistic competitive firm $y_{j,t} = Z_t l_{j,t} - \Phi$, where Φ is an Overhead Cost. They hire labor at real wage w_t and set a constant markup of $\mu = \frac{\varepsilon}{\varepsilon-1}$. Hence, $P_t = \mu W_t / Z_t$, and real wage equals $w_t = Z_t / \mu$ at

every date in steady-state. There is a mass 1 of outstanding firm shares with end-of-period price p_t^e (Auclert, Rognlie, and Straub 2025).

We assume sticky prices following Rotemberg (1982), standard in Neo-Keynesian's literature, with a quadratic adjustment cost

$$\Psi_t(\pi_t) = \frac{\mu}{\mu - 1}\frac{1}{2\kappa}\left[\log(1+\pi_t)\right]^2 Y_t.$$

Firms' dividends are taxed at rate τ_d, so $d_t = (1-\tau_d)(Y_t - w_t L_t - \Phi) - \Psi_t(\pi_t)$, where $L_t = \mathbb{E}[l_{j,t}]$ is the aggregate formal labor. Such as Hagedorn, Manovskii, and Mitman (2019), the Phillips curve for price inflation is given by

$$\ln(1+\pi_t) = \kappa\left[\frac{w_t}{Z_t} - \frac{1}{\mu}\right] + \frac{\ln(1+\pi_{t+1})}{1+r_{t+1}} \cdot \frac{Y_{t+1}}{Y_t},$$

Additionally, there is a representative informal sector operating with flexible prices and decreasing returns to scale, $Y_t^I = Z_t^I \left(N_t^I\right)^{\alpha_I}$, with $\alpha_I < 1$. The informal good is a perfect substitute for the formal one, so it sells at the same price P_t. Informal units are self-employed or revenue-sharing producers, so the entire product is distributed among their workers. The informal labor demand is

$$w_t^I = \frac{Y_t^I}{N_t^I} = Z_t^I \left(L_t^I\right)^{\alpha_I - 1}.$$

Also, due to decreasing returns to scale, an extra worker congests the average product, so the marginal revenue is given by $\partial\left(w_t^I N_t^I\right)/\partial N_t^I = \alpha_I w_t^I$, where α_I acts as a second markup for the informal market. Because the informal wage is also the most flexible nominal wage in the economy, the informal sector absorbs the employment adjustment while its wage absorbs the price adjustment, which makes informal income the most volatile in a recession.

Unions

There are two unions in this economy, one for each sector. Nominal wages are assumed to be sticky as in Erceg, Henderson, and Levin (2000), where unions set nominal wages to maximize agent utility subject to Rotemberg (1982)'s adjustment costs. We assume that unions allocate all labor hours uniformly across agents, so $h_{i,t} = h_t$ (Auclert, Rognlie, and Straub 2025).

Also, as in Hagedorn, Manovskii, and Mitman (2019), union's objective is to maximize the utility of an agent with average consumption $\bar{C}_t = \int_0^1 c_{it}\, di$, discounted by the average discount factor $\bar{\beta}$ (Auclert, Rognlie, Souchier, et al. 2021). This results in two Phillips

curves for wage inflation $\pi_t^w = (1+\pi_t)w_t/w_{t-1}$ and $\pi_t^{w_I} = (1+\pi_t)w_t^I/w_{t-1}^I$ given by

$$\ln(1+\pi_t^w) = \kappa_w \left[\psi \cdot h_t^{1/\varphi} - \frac{(1-\tau_\ell)w_t}{(1-\tau_t^c)\mu_w F_t}\int_F e_{i,t}\theta_{i,t}u'(c_{i,t})d\Lambda_t\right] + \bar{\beta}_F \ln(1+\pi_{t+1}^w),$$

and

$$\ln(1+\pi_t^{w_I}) = \kappa_w^I \left[\psi \cdot \left(h_t^I\right)^{1/\varphi} - \frac{\alpha_I w_t^I}{(1-\tau_t^c)\mu_w I_t}\int_I e_{i,t}\theta_{i,t}u'(c_{i,t})d\Lambda_t\right] + \bar{\beta}_I \ln(1+\pi_{t+1}^{w_I}),$$

where the informal adjustment cost is lower than the formal one, so $\kappa_w^I > \kappa_w$. The informal sector is the most flexible in this economy, which translates into greater instability during periods of crisis.

The Phillips Curve calculations for price and wage are found in the Appendix.

Government

Monetary Policy follows a standard lagged Taylor Rule, without microfoundations and only with an inflation target, $i_t = r_t^* + \phi\pi_t$, where r^* is the neutral real interest rate, i_t is the nominal interest rates, pi_t is the inflation, and $\phi > 1$ (Taylor 1993). Let $1 + r_t = (1+i_{t-1})/(1+\pi_t)$ be the real interest rates.

In the governmental sphere, we have a government that collects taxes, makes transfers, purchases public goods, and incurs debt; therefore, the government's budget constraint is characterized by

$$G_t + T_t \cdot \mathrm{BF}_t + (1+r_t)B_{t-1} = B_t + \tau_\ell w_t L_t + \tau_t^c C_t + \tau_d(Y_t - w_t L_t).$$

We consider two types of government policy, a revenue-neutral and a deficit-financed. The policymaker chooses a path of the conditional transfers and adjusts taxes or debt such that the budget is balanced at each period.

The revenue-neutral policy sets a path of T_t and adjusts taxes such that

$$\tau_t^c = \frac{r_t\bar{B} + G_t + T_t \cdot \mathrm{BF}_t + (\tau_d - \tau_\ell)w_t L_t - \tau_d Y_t}{C_t},$$

which means that the fiscal policy is passive and the monetary policy is active. The nominal interest rate adjust at least one-for-one to changes in inflation.

However, governments normally finance themselves by increasing deficit and issuing new debt. The deficit-financed policy sets a path of T_t and τ_t^c, adjusting the debt to pay off. We assume that the government repays its debt in the long run by increasing taxes, with elasticity ϕ_B,

$$\tau_t^c = \bar{\tau}^c + \phi_B \frac{B_{t-1} - \bar{B}}{C_t}.$$

Or, in another words, using Auclert, Rognlie, and Straub (2025) notation, the debt persistency is defined by

$$dB_t = (1 + r_t)dB_{t-1} - C_t d\tau_t^c = (1 + r - \phi_B)dB_{t-1} = \rho_B dB_{t-1},$$

where ρ_B is the debt persistency, which we calibrate so that debt has a half-life of 10 years, so $\rho_B = 0.5^{1/40} \approx 0.983$ and $\phi_B = 0.030$.

This situation leads to larger multipliers and great inequality (Auclert, Rognlie, and Straub 2023). In the presence of nominal rigidity and non-Ricardian consumer behavior, Angeletos, Lian, and Wolf (2024) show that fiscal deficits of this sort self finance, so $\lim_{t\to\infty} B_t = \bar{B}$.

Finally, there is a mutual fund collecting household savings $a_{i,t}$ and investing on government bonds and the stock market. No-arbitrage condition garanties that

$$(1 + r_{t+1})p_t^e = p_{t+1}^e + d_{t+1} \quad \Leftrightarrow \quad p_t^e = \sum_{s\geq 1} \frac{d_{t+s}}{\prod_{k=1}^{s}(1 + r_{t+k})},$$

which implies

$$1 + r_t^a = \frac{(1 + r_t)B_{t-1} + d_t + p_t^e}{B_{t-1} + p_{t-1}^e} = 1 + r_t, \quad \forall t \geq 1.$$

At $t = 0$ the shock is unanticipated, so households hold a portfolio bought at steady-state prices. The initial return r_0^a is such that the liquidation value of the mutual fund equals the beginning-of-period value of its stock and bond holdings,

$$(1 + r_0^a)A = p_0^e + d_0 + (1 + r_0)B,$$

so $1 + r_0^a$ is a weighted average of the initial stock return $(p_0^e + d_0)/p^e$ and the bond return $1 + r_0$, with weights given by the steady-state portfolio shares p^e/A and B/A. The shock thus revalues household wealth on impact through both assets, while every Euler equation prices the ex-ante return $r_{t+1}^a = r_{t+1}$.

Market Clearing

Finally, the asset, labor, and goods markets reach equilibrium (Market Clearing Conditions). First, the asset market reaches equilibrium when household savings equal stocks and government bonds

$$A_t = \int a_{i,t}\, d\Lambda_t(s, b, \theta, \beta, e, a) = B_t + p_t^e;$$

the labor market clears when the share of employment in the workforce equals the firm's demand for labor

$$L_t = \int \mathbb{1}\{s_{i,t} = F\} \cdot \theta^F_{i,t} \cdot e_{i,t} \cdot h_{i,t} \, d\Lambda_t(.);$$
$$L^I_t = \int \mathbb{1}\{s_{i,t} = I\} \cdot \theta^I_{i,t} \cdot e_{i,t} \cdot h_{i,t} \, d\Lambda_t(.);$$

and the goods market clear when $Y + Y^I + y^U_t \alpha_U = C + G + \Psi_t$. We set the goods market to clear by Walras' Law. Lastly, the BF benefit is given to all eligible households,

$$\text{BF}_t = \int b_{i,t} \, d\Lambda_t(.).$$

Equilibrium. Given initial conditions $(B_{-1}, p^e_{-1}, \Lambda_0)$ and exogenous paths $\{T_t, r^*_t\}_{t\geq 0}$, an equilibrium is a sequence of prices $\{w_t, w^I_t, \pi_t, \pi^w_t, \pi^{w_I}_t, i_t, r_t, r^a_t, p^e_t\}$, macroeconomic aggregates $\{Y_t, Y^I_t, L_t, L^I_t, h_t, h^I_t, d_t, B_t, C_t, A_t, G_t, \text{BF}_t\}$, transition matrices $\{\Pi_t\}$, household policies $\{c_t, a_t\}$ over the state $(s, b, \theta, \beta, e, a)$, and distributions $\{\Lambda_t\}$ such that, for all $t \geq 0$:

1. The policies solve the Bellman equations of each sector;
2. Firms price according to the Price Phillips curves and the union sets wages according to the Wage Phillips curves, with $h^s_{i,t} = h^s_t$ for every worker;
3. The government satisfies its budget constraint and fiscal rule, and monetary policy follows the Taylor rule, with $1 + r_t = (1 + i_{t-1})/(1 + \pi_t)$; The mutual fund prices stocks by no-arbitrage and pays r^a_t;
4. $A_t = B_t + p^e_t$, L_t and L^I_t equal the effective labor supplied by formal and informal workers, $\text{BF}_t = \int b \, d\Lambda_t$, and $Y_t + Y^I_t = C_t + G_t + \Psi_t$ holds by Walras' law.

A steady state is an equilibrium in which prices and aggregates are constant, Λ is invariant, and Π is a fixed point. Numerically, the transition solves for $\{B_t, L_t, h_t, h^I_t, \pi_t, w_t, w^I_t, \tau^\ell_t, p^e_t\}$ such that the fiscal rule, the asset and labor markets, the informal labor demand, the three Phillips curves, the government budget and the no-arbitrage condition hold.

2. Calibration

Data

To calibrate this model, we use Brazilian data and standard parameters in HANK Literature. Most of the data refers to their average values in 2025.

The labor market moments were obtained in *Pesquisa Nacional por Amostra de Domicílio Contínua* (PNAD-C), an annual survey reporting formality status, wages, weekly worked hours, etc. This panel is interesting to match the sector transitions and

the worked hours and wages, for both formal and informal households. However, since it is a survey, it might have under-reporting of informal households.

Some general macroeconomic measures were also used, such as the Government Debt (in the Brazilian National Treasure), the interest rates, and the inflation, available in the *Índice Brasileiro de Geografia e Estatística* (IBGE). The Gini coefficient of wealth was obtained from the 2025 Global Wealth Report, from UBS, and the Wealth Inequality Statistics were obtained from the 2024 Wealth Inequality Database[1]. *Bolsa Família* data is available in *Cadastro Único* (CadÚnico), in VIS Data[2], a platform with several statistics about the benefit.

Additionally, Annual PNAD-C has information about whether a individual is beneficiary of the benefit. We identify each household and consider every individual living in a beneficiary household as receiving the benefit. Among those, we calculate the mean reported benefit T and the mean share of beneficiaries.

Table 1: Bolsa Família 2025: Statistics

	PNAD	VisData	Ratio
Households (M)	13.64	18.71	1.37
Individuals (M)	46.60	48.92	1.05
People/HH	3.42	2.61	0.77
Value (R$/month)	669.55	710.65	1.06

Table 2: Bolsa Família 2025: Coverage

	%
BF / Labor Force	16.1
BF in Formal	7.2
BF in Informal	23.9
BF in Unemployed	33.6
BF in Non-Participating	21.8
Spending (Labor Force) / Wage Bill	2.1
$T/\mathbb{E}(y^F)$	10.0
$\bar{y}_{\text{law}}/\mathbb{E}(y^F)$	17.6

The rule to receive BF is based on a $y_{\text{law}} = R\$\,218$ threshold for the family income per capita, therefore there are some formal individuals receiving the benefit. Also, CadÚnico normally delay to remove the benefit for non-eligible individuals, which can explain why some formal individuals receive BF.

In order to construct quarter-to-quarter transition probabilities across the three sectors $s \in \{F, I, U\}$, we reconstruct the PNAD-C rotating panel using Osório (2022), which

[1] Available in WID World.

[2] Available in VIS DATA 3 and Cadastro Único.

estimates the dwelling and individual identifiers in the survey. The estimation of the pandemic years is not reliable.

The formality status is defined as (i) formal workers have signed work card, are militaries, or employers, and (ii) informal workers do not have a signed work card, are self-employed, or unpaid family. We only consider household in the labor force, dropping those who are non-participating, retired, or students, to keep the empirical states aligned with the model's. To maintain the internal consistency with the model, we use hourly wage moments, which help to reduce the part-time informal workers who earn less.

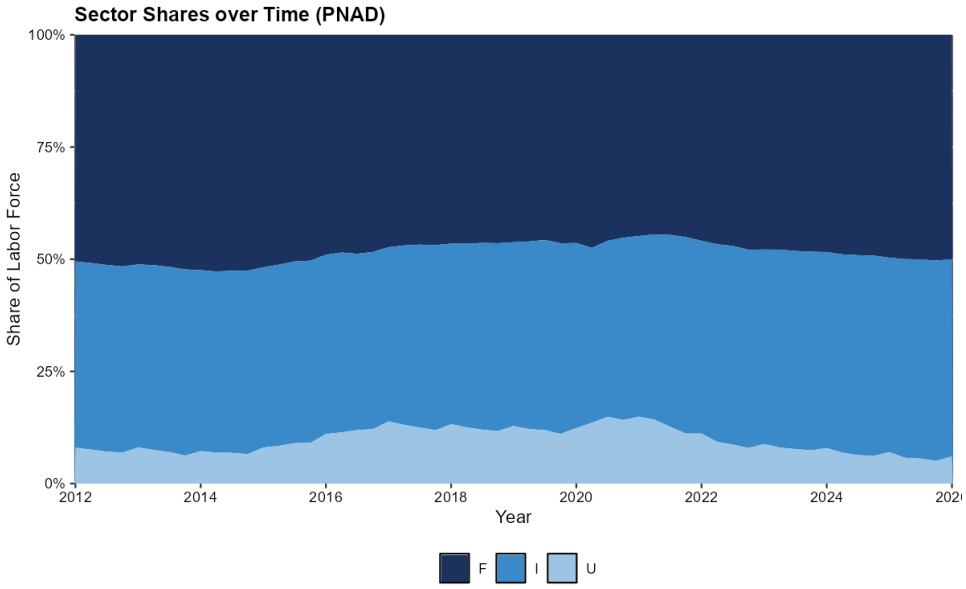


Figure 1: Share of Formal, Informal, and Unemployed Households

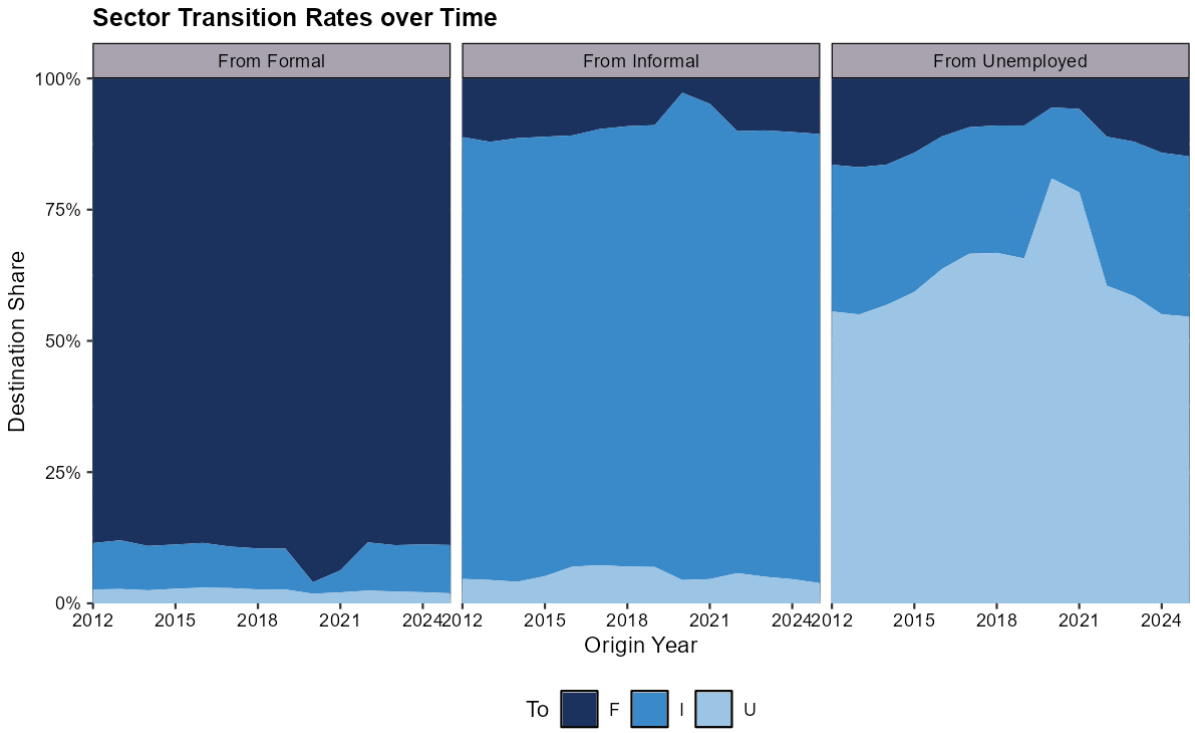


Figure 2: Sector Transitions over time

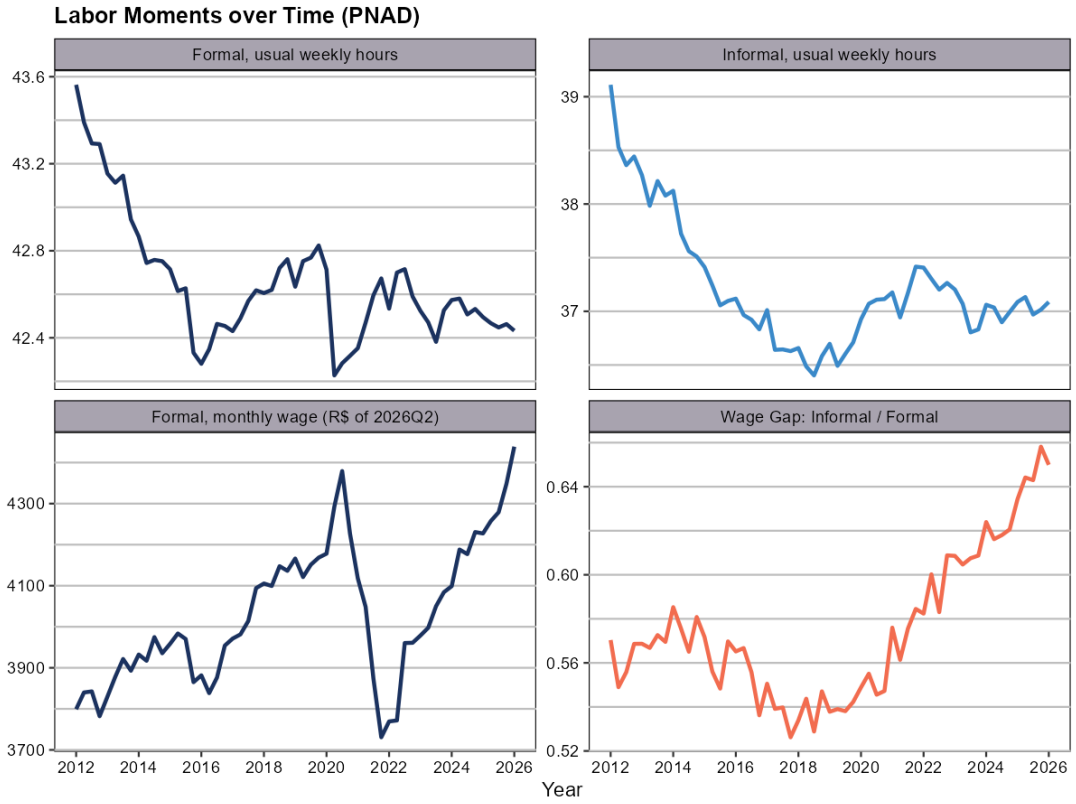


Figure 3: Wage and Worked Hours for each Sector



Figure 4: Average Worked Hours by Wage for each Sector

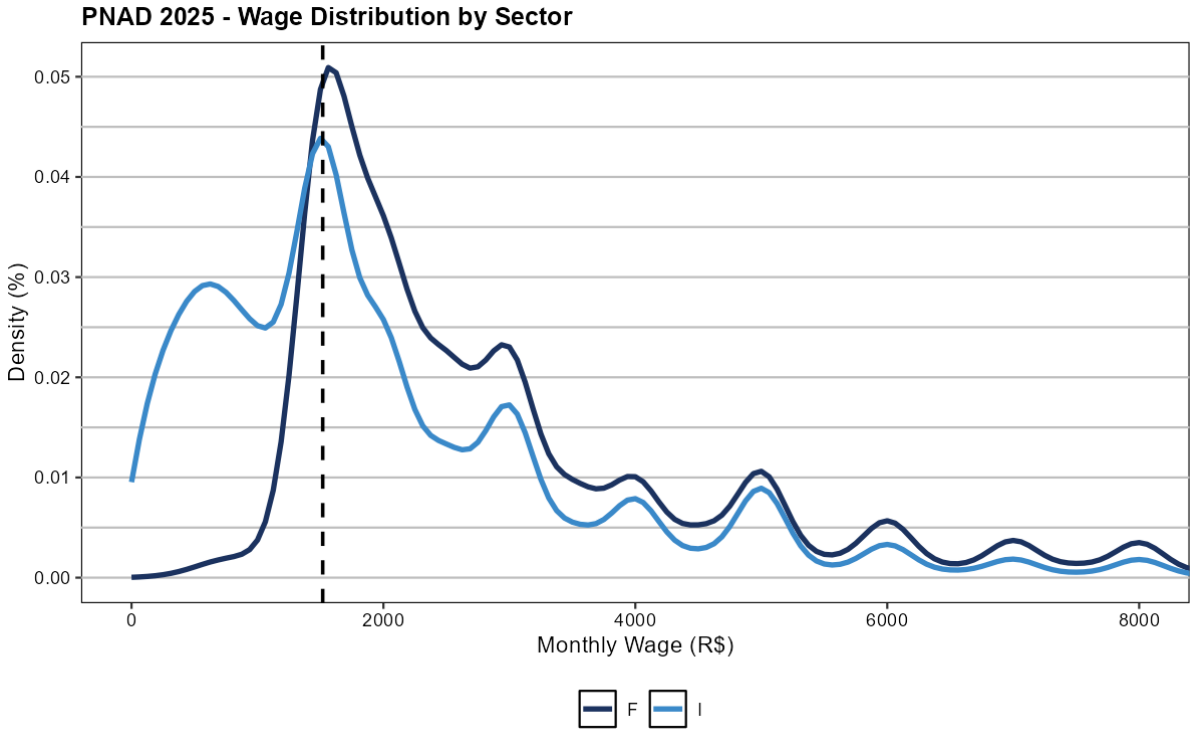


Figure 5: Wage Distribution for each Sector

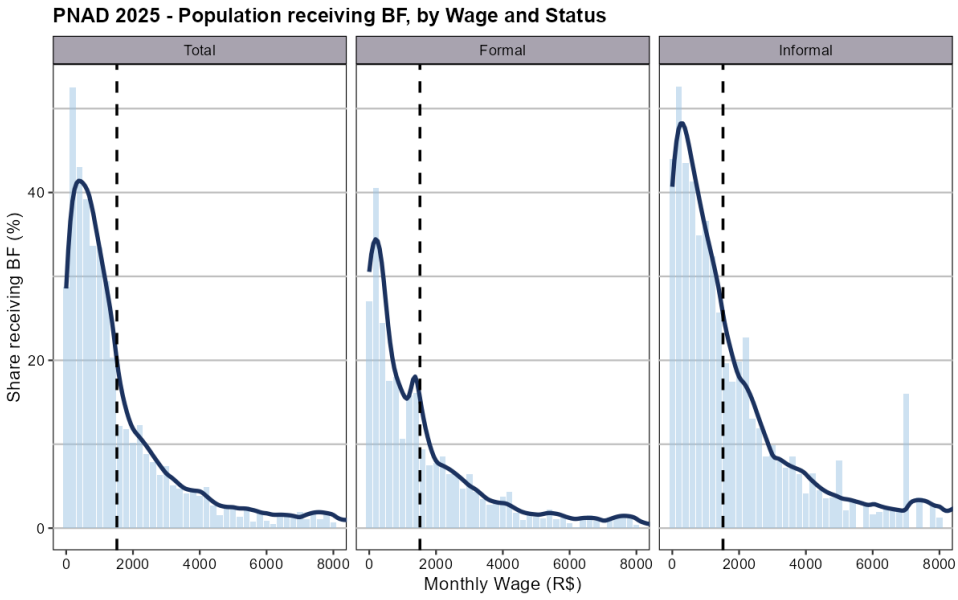


Figure 6: Share of Households Receiving Benefit by Wage Level

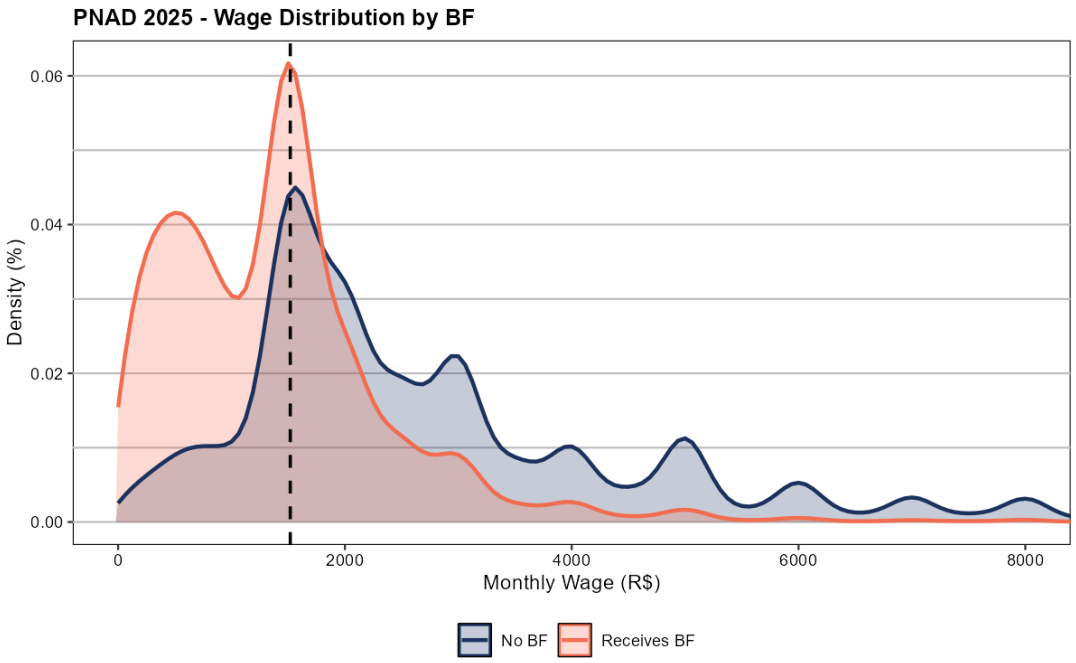


Figure 7: Wage Distribution by Benefit Status

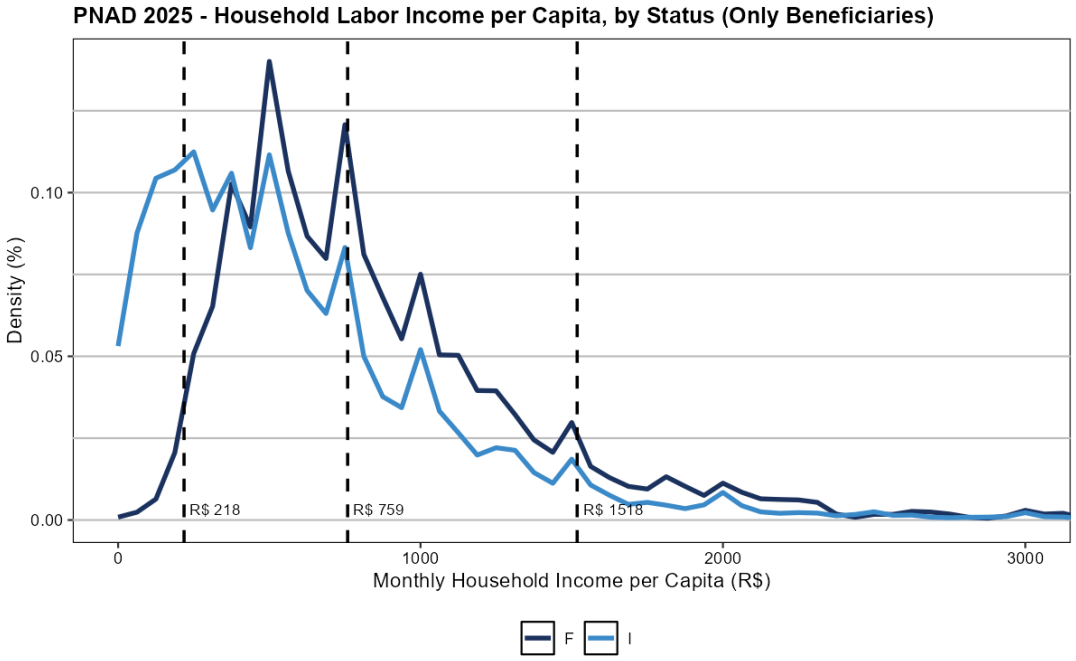


Figure 8: Familiar Wage Distribution by Sector, for Beneficiaries

Methods

The main computational methods are the Sequence-Space Jacobian (SSJ) (Auclert, Bardóczy, et al. 2021), and the Endogenous Grid Method (EGM) (Carroll 2006). The EGM was combined with the discrete choice among sectors using Iskhakov et al. (2017), with taste shocks.

The central idea of SSJ is to summarize the entire model by a matrix equation $\mathbf{F}_t(\mathbf{X}, \mathbf{Z}) = \mathbf{0}$, where $\mathbf{X}$ are the endogenous variables and $\mathbf{Z}$ are the exogenous ones. The Implicit Function Theorem allows us to differentiate this system and obtain General Equilibrium Jacobians, written as

$$d\mathbf{X} = \mathbf{G}\, d\mathbf{Z} = -\mathbf{F}_{\mathbf{X}}^{-1}\mathbf{F}_{\mathbf{Z}}\, d\mathbf{Z}.$$

For each shock in *Bolsa Família*, we can compute the Jacobian $dC = \mathbf{G}_{C,T}\, dT$ of the change in the conditional transfer on household consumption. The Jacobian is computed by perturbing the exogenous variable T and solving the model for the new equilibrium, then calculating the change in consumption C.

Further, we can compute the consumption response to a monetary shock $dC = \mathbf{G}_{C,i}\, di$ in the presence of the transfer, and compare it to the counterfactual. We measure the stabilization effect of the transfer as the difference between the response in the economy

with the program and in the $T = 0$ counterfactual,

$$\text{Stab}(T) = dC\Big|_{T>0} - dC\Big|_{T=0},$$

read either as an impulse response or as the induced change in the volatility of consumption $\Delta\text{Var}(C_t)$, a program that smooths fluctuations delivers $\Delta\text{Var}(C_t) < 0$. This effect can be decomposed in a insurance and a composition channel,

$$d\mathbf{X} = d\mathbf{X}^{\text{ins}} + \left(d\mathbf{X} - d\mathbf{X}^{\text{ins}}\right) = \mathbf{G}^{\text{ins}}\, d\mathbf{Z} + \left(\mathbf{G} - \mathbf{G}^{\text{ins}}\right) d\mathbf{Z}.$$

The sectoral block of the transition matrix Π_t is endogenous and depends on the continuation values, $\Pi_t = \Pi(\mathbf{V}_t, \Lambda_t)$, where $\mathbf{V}_t = (V_t^F, V_t^I, V_t^U)$. The household's distribution evolves by combining the asset policy with the Markov transition over the states, $\Lambda_{t+1} = \mathcal{T}(\Pi_t, a_t(s, \theta, \beta, e))$. To solve the steady-state distribution, we iterate until Π_t converges to its fixed point, analogue to Aguirregabiria and Mira (2007).

The insurance channel is captured by the frozen-Π. Holding sectoral sorting fixed, the transfer relaxes the budget of unemployed and low-income informal households, and, since these are high-MPC, it dampens the aggregate response (Auclert, Rognlie, and Straub 2024).

The composition channel is captured by the moving-Π Jacobian, because the CCT raises the value of informality relative to formal work, so workers are tilted toward the informal sector, which is riskier. The insurance term is stabilizing, while the composition is potentially destabilizing. The sign and magnitude of the net effect are the quantitative answer to our question.

Letting the transition respond, $d\Pi_t = \Pi_{\mathbf{V}}\, d\mathbf{V}_t$, yields the general equilibrium Jacobian $\mathbf{G}$. This dynamic is obtained as the dynamic analogue of the steady-state: given a shock path, we iterate so that households' sectoral choices are consistent with the values those very choices generate along the transition (Auclert, Bardóczy, et al. 2021).

Calibration

Each model period stands for one quarter and the computational time horizon is 100 quarters (25 years). We internally calibrate the steady state by setting β_H to balance the asset market; π_F, π_I, π_{UF}, π_{UI} to solve the sectoral transition; and λ, $\bar{y}$, ρ_F to solve the BF coverage by sector in steady state (Table 2). The external calibration is shown in table 3 and the endogenous calibration is in 4.

The idiosyncratic productivity follows a log-normal $AR(1)$ process, $\ln e' = \rho_e \ln e + \varepsilon^e$, with $\varepsilon^e \sim N(0, \text{Var}(\varepsilon^e))$ and $\text{Var}(e) = \sigma_e^2$. This is approximated by a finite Markov chain with $N_e = 15$ grid points (Rouwenhorst 1995). We also assume a log-spaced grid for assets $[0, 100]$ with $N_a = 100$ points, sufficient to cover the bottom and the top of distribution,

Table 3: External Calibration

	Description	Value	Source
Households and Labor Market			
γ	EIS	0.5	Standard in Literature
φ	Informal Frisch Elasticity	0.5	Standard in Literature
ψ	Disutility of Labor	0.855	Normalization: $h_F = 1.0$
β_H	Patient's Discount Factor	0.956	$A = B + p^e$
A/Y	Households' Liquid Assets	0.640	ANBIMA
δ_F	Job-Loss Probability to Formal	0.019	PNAD-C Panel (Osório 2022)
δ_I	Job-Loss Probability to Informal	0.038	PNAD-C Panel (Osório 2022)
σ_S	Job Acceptance Smoothness	0.600	Internal Consistency
Firms			
α_I	Informal Returns to Scale	0.85	Standard
w, w^I	Average Wage	1.0	Normalization, Reference
μ, μ_w	Price Markup	1.11	Basu and Fernald (1997)
κ	Price-NKPC Slope	0.10	Kaplan, Moll, and Violante (2018)
κ_w	Formal Wage-NKPC slope	0.62	Caccioli (2026)
κ_w^I	Informal Wage-NKPC slope	1.869	Three times higher than κ_w
Φ	Overhead Cost to Firms	0.098	Calibrate p^e, from ANBIMA
Government and Monetary Policy			
ϕ	Taylor Rule Coefficient	1.5	Standard (Taylor 1993)
r^*	Neutral Real Interest Rate	1.2%	5% Annual Average
τ_ℓ	Labor Income Tax Rate	13%	Tax Payroll for Average Worker
τ^c	Consumption Tax Rate	21%	National Treasury
τ_d	Profit Tax Rate	34%	IRPJ 25%, and CSLL 9%
ϕ_B	Sensitivity to Pay the Debt	0.030	B half-life of 10 years
$BF \cdot T/W$	BF Spending (%Wage Bill)	2.1%	CadÚnico, PNAD-C
ρ_I	Informal Fiscalization Prob.	0.12	CadÚnico, 2y Review

Table 4: Endogenous Calibration

	Description	Value	Target	Model	Data	Source
Simulated Methods of Moments (SMM)						
μ_I	Informal Average Productivity	0.036	Δ^y_{FI}	−0.014	−0.030	PNAD-C
σ_e	Productivity Std. Dev.	0.629	ξ	0.594	0.748	PNAD-C
σ_F	Formal Std. Dev. Produc.	0.422	σ^y_F	0.761	0.723	PNAD-C
σ_I	Informal Std. Dev. Produc.	0.303	σ^y_I	0.747	0.894	PNAD-C
ρ_e	Persistence of Productivity	0.956	ρ^y_4	0.799	0.828	PNAD-C Panel
β_L	Impatient's Discount Factor	0.664	MPC	0.27	0.20	Galvêas (2026)
ω_I	Share of Impatient Agents	50%	Gini	0.692	0.82	UBS Global Report
Internal Consistency						
π_F	Formal Offer Probability	0.528	$I \to F$	10.6%	10.6%	PNAD-C Panel
π_F^U	Formal Offer Prob. (Unemp.)	0.299	$U \to F$	14.9%	14.9%	PNAD-C Panel
π_I	Informal Offer Probability	0.640	$F \to I$	9.3%	9.3%	PNAD-C Panel
π_I^U	Informal Offer Prob. (Unemp.)	0.519	$U \to I$	31.2%	31.2%	PNAD-C Panel
λ	BF Entry Probability	0.20	BF_U	33.6%	33.6%	PNAD-C
$\bar{y}$	BF Threshold for Informals	0.52	BF_I	23.9%	23.9%	PNAD-C
ρ_F	Formal Fiscalization Prob.	0.24	BF_F	7.2%	7.2%	PNAD-C

and a log-normal sector productivity grid $\ln \theta^s \sim N(\mu_s - \sigma_s^2/2, \sigma_s^2)$, that is discretized by Equiprobable Bins with $N_\theta = 7$ nodes, where $\mu_s = \log \mathbb{E}\left[\theta^s_{i,t}\right]$ represents the worker's comparative advantage.

In the household preferences, elasticity of intertemporal substitution $\gamma = 0.5$ and Frisch elasticity $\varphi = 0.5$ are standard in literature[3]. The disutility of labor $\psi = 0.855$ is set to normalize worked hours $h = 1.0$ in steady state. Each household has a $q = 0.01$ probability of drawing a new discount factor (β_L, β_H), corresponding to an average type duration of 25 years, used to capture a generation (Auclert, Rognlie, and Straub 2025). We let y_U be 2% of average formal wage.

The sector transition probabilities in PNAD-C are calibrated by the layoff probabilities, $\delta_F = 0.019$ and $\delta_I = 0.038$, and by the job offer probabilities $(\pi_F, \pi_F^U, \pi_I, \pi_I^U) = (0.528, 0.299, 0.640, 0.519)$.

Table 5: Quarterly Labor-Market Transitions (%)

	Model			**PNAD-C**		
Origin	*F*	*I*	*U*	*F*	*I*	*U*
Formal	88.81	9.32	1.87	88.81	9.32	1.87
Informal	10.64	85.61	3.75	10.64	85.61	3.75
Unemployed	14.93	31.25	53.82	14.93	31.25	53.82
Stationary Share	49.84	44.52	5.63	49.84	44.52	5.63

The average wage is normalized $w = w^I = 1.0$, so that we can reference the model as a percentage of the Wage Bill. We defined $\alpha_I = 0.85$ and $\mu = 1.11$, which is standard in the HANK literature (Kaplan, Moll, and Violante 2018; Auclert, Rognlie, and Straub 2025)[4], $\kappa = 0.10$, similar to Kaplan, Moll, and Violante (2018), and $\kappa_w = 0.62$, consistent with Caccioli (2026) estimations[5]. We set $\Phi = 0.098$ so that $p^e = d/r = 0.498$ in steady state and β_H so that $A/Y = B/Y + p^e/Y = 0.640$, from ANBIMA data.

Taylor Rule is pretty standard, with $\phi = 1.5$ and $r^* = 1.2\%$, consistent with a 5% annual real interest rate for Brazilian economy (Santos and Muinhos 2024). Also, we set $\phi_B = 0.030$ so that the half-life of the debt is 10 years, slower than Auclert, Rognlie, and Straub (2025) and consistent with Campos and Cysne (2019). Wage tax is based on the average formal worker contribution, $\tau_\ell = 13\%$, profit's tax is based on the statutory rates for Brazil, IRPJ and CSLL, which are 25% and 9% respectively, and consumption tax is based on taxes on goods and services, $\tau^c = 21\%$, from Brazilian National Treasury.

We calibrated the *Bolsa Família* transfer size $T = 0.198$ to match the *Bolsa Família*

[3]EIS $\gamma = 0.5$ is consistent with micro estimates surveyed by Havranek et al. (2015).

[4]Markup $\mu = 1.11$ is also consistent with the estimates of Basu and Fernald (1997) for US. De Loecker, Eeckhout, and Mongey (2021) show that the Brazilian markup is similar to the US.

[5]We calculate κ_w following the standard Calvo formula $\kappa_w = \frac{(1+\theta)(1-\tilde{\beta}\theta)}{\theta}\frac{1}{1+\varepsilon_w\varphi}$, with the quarterly wage adjustment probability $1-\theta = 25.4\%$ from Caccioli (2026).

expenditure as a percentage of the Wage Bill, $\text{BF} \cdot T/W$. In order to keep the internal consistency of the model, we consider T as the average transfer received only by indivuals in labor force, disconsidering children and elderly.

The audit probability for informals $\rho_I = 0.12$ is set by an average 2 years time for review, by the CadÚnico regulation. While the entry probability $\lambda = 0.20$, the threshold $\bar{y}/y^F = 0.52$, and the audit probability for formals $\rho_F = 0.24$ are internally calibrated to match the coverage by sector (Table 2). $\bar{y}$ value is similar to the $35th$ percentile of formal income (Ferriere and Navarro 2025), $y_{35}/y^F = 0.359$, and ρ_F is consistent with an average time of 12 months to lose the benefit.

Finally, we internally calibrate parameters using the Simulated Method of Moments (SMM) by a Nelder-Mead algorithm, explained in detail in the appendix. We jointly estimate $(\mu_I, \sigma_F, \sigma_I, \sigma_e, \rho_e, \beta_L, \omega_I)$ to match the average wage gap between sectors $\xi = \mathbb{E}[y_F]/\mathbb{E}[y_I]$, the standard deviation of each sector's income, the autocorrelation of wages $\rho_4^y := \text{Corr}(\ln y_t, \ln y_{t-4})$, the wage difference of switchers $\mathbb{E}[\Delta \ln y_t | F \to I] = \Delta_{FI}^y$ and $\mathbb{E}[\Delta \ln y_t | I \to F] = \Delta_{IF}^y$, the Wealth Gini Coefficient (0.82, from UBS Global Report), the income-weighted Marginal Propensity of Consumption (MPC)[6] (0.20) and the share of Hand-to-Mouth households (35.0%), letting the Top 10% Share of Wealth (71.9%, from World Inequality Database), amonth other statistics untargeted.

[6]Income-Weighted MPC is consistent with the estimates of Costa-Junior, Gadelha, and Garcia-Cintado (2025) and Galvêas (2026).

Table 6: Targeted and Untargeted Moments

Moment	Data	Model	Gap
Wealth Gini	0.820	0.693	-0.127
Hand-to-Mouth	0.350	0.337	-0.013
Aggregate MPC	0.200	0.276	+0.076
ξ, hourly wage gap	0.748	0.889	+0.141
$\text{sd}(\log y_F)$	0.723	0.693	-0.030
$\text{sd}(\log y_I)$	0.894	0.798	-0.096
$\mathbb{E}[\Delta \log y \mid F \to I]$	-0.030	-0.041	-0.011
$\mathbb{E}[\Delta \log y \mid I \to F]$	0.079	0.158	+0.079
ρ_4^y	0.828	0.761	-0.066
Wealth, Bottom 50%	0.020	0.058	+0.038
Wealth, Top 10%	0.719	0.524	-0.195
Wealth, Top 1%	0.395	0.118	-0.277
share_F	0.498	0.498	+0.000
share_I	0.445	0.445	-0.000
share_U	0.056	0.056	-0.000
q_{10}^F	-1.057	-1.137	-0.080
q_{25}^F	-0.868	-0.467	+0.401
q_{50}^F	-0.463	-0.127	+0.336
q_{75}^F	0.048	0.209	+0.161
q_{90}^F	0.700	0.545	-0.155
q_{10}^I	-1.736	-1.510	+0.226
q_{25}^I	-1.144	-0.916	+0.228
q_{50}^I	-0.686	-0.285	+0.401
q_{75}^I	-0.216	0.092	+0.308
q_{90}^I	0.413	0.520	+0.107
$\rho_1(\log y)$	0.864	0.938	+0.073
h^I/h^F	0.973	0.942	-0.031
$T/\mathbb{E}[y^F]$	0.100	0.112	+0.011
$P(\Delta \log y < 0 \mid F \to I)$	0.450	0.647	+0.197
$P(\Delta \log y < 0 \mid I \to F)$	0.381	0.272	-0.110

3. Results

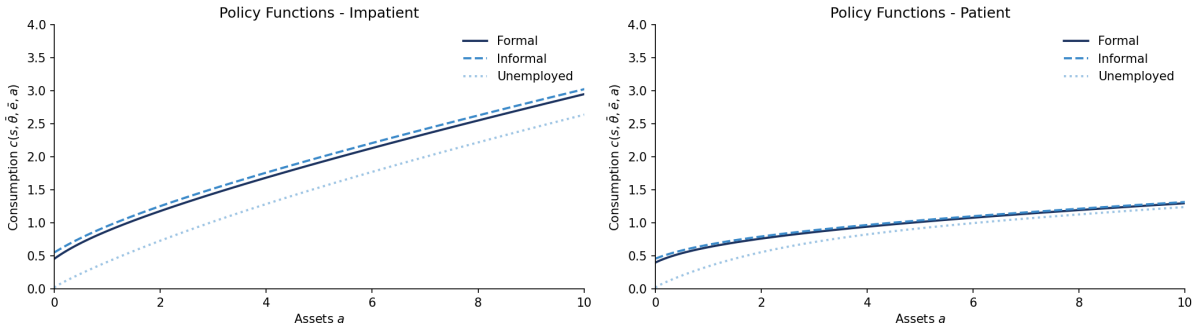


Figure 9: Steady-State Consumption Policy Function

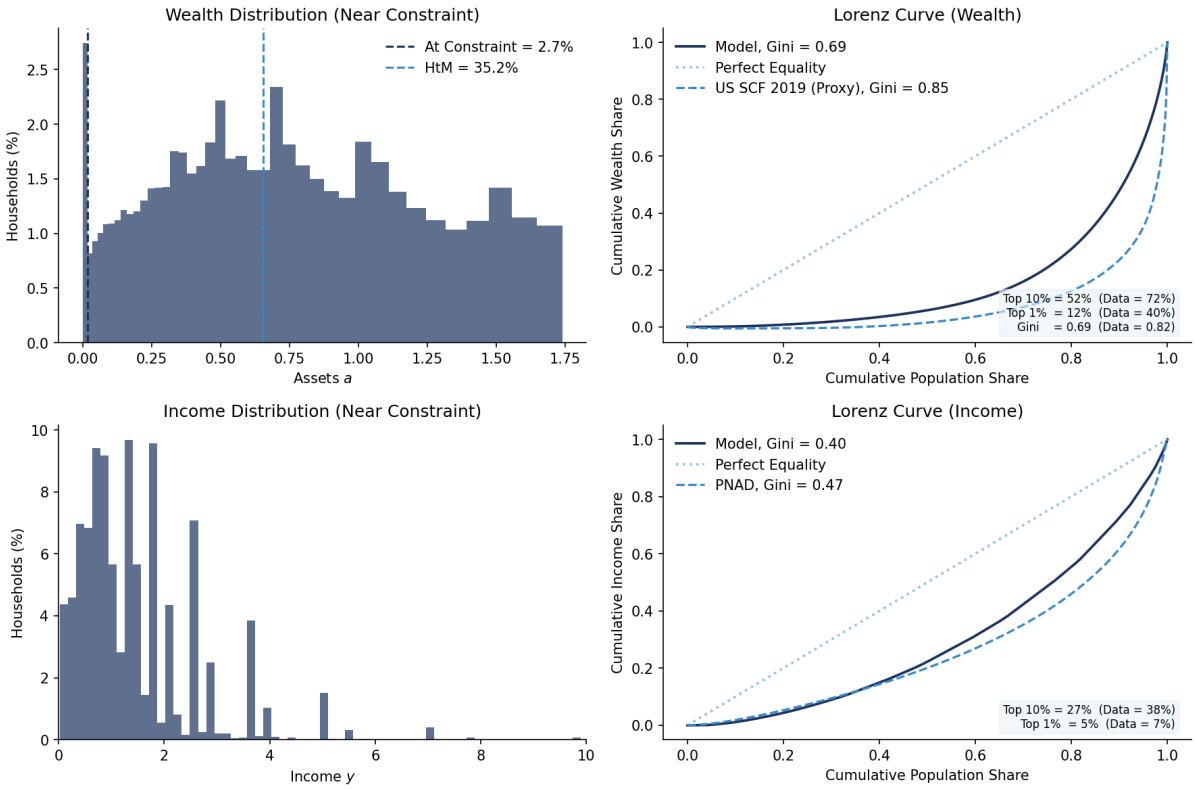


Figure 10: Wealth Distribution

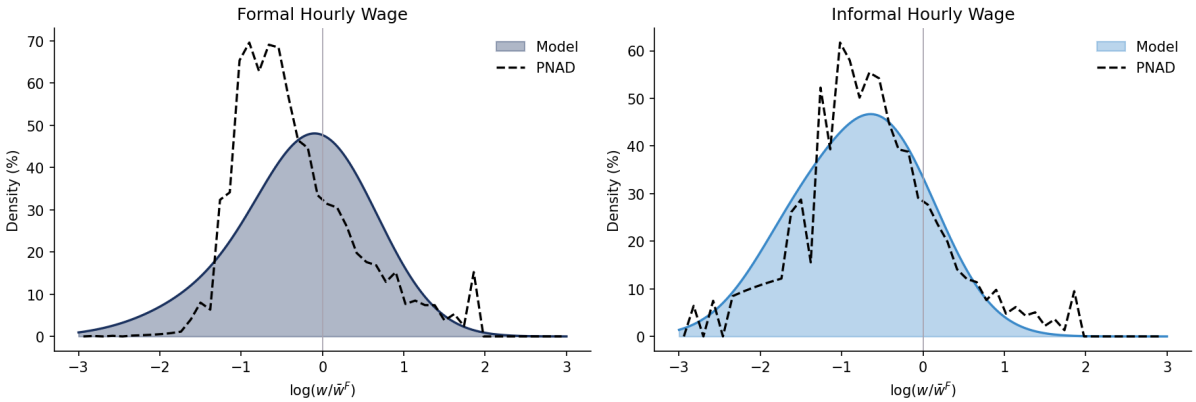


Figure 11: Income Distribution

Table 7: Steady State: Bolsa Família vs. No-Transfer Counterfactual

	No BF	With BF	Δ
Formal Share α_F	55.5%	49.8%	-5.7
Informal Share α_I	39.2%	44.5%	+5.3
Unemployment α_U	5.3%	5.6%	+0.4
Hand-to-Mouth	30.1%	35.2%	+5.1
Wealth Gini	0.671	0.692	+0.021
Wealth Share, Bottom 50%	6.7%	5.8%	-0.8
Wealth Share, Top 10%	49.7%	52.1%	+2.4
Consumption C	1.305	1.242	-0.064
BF Spending / GDP	0.0%	2.1%	+2.1
Welfare $\mathbb{E}[V]$, Δ = CEV	-18.532	-18.045	+3.43%

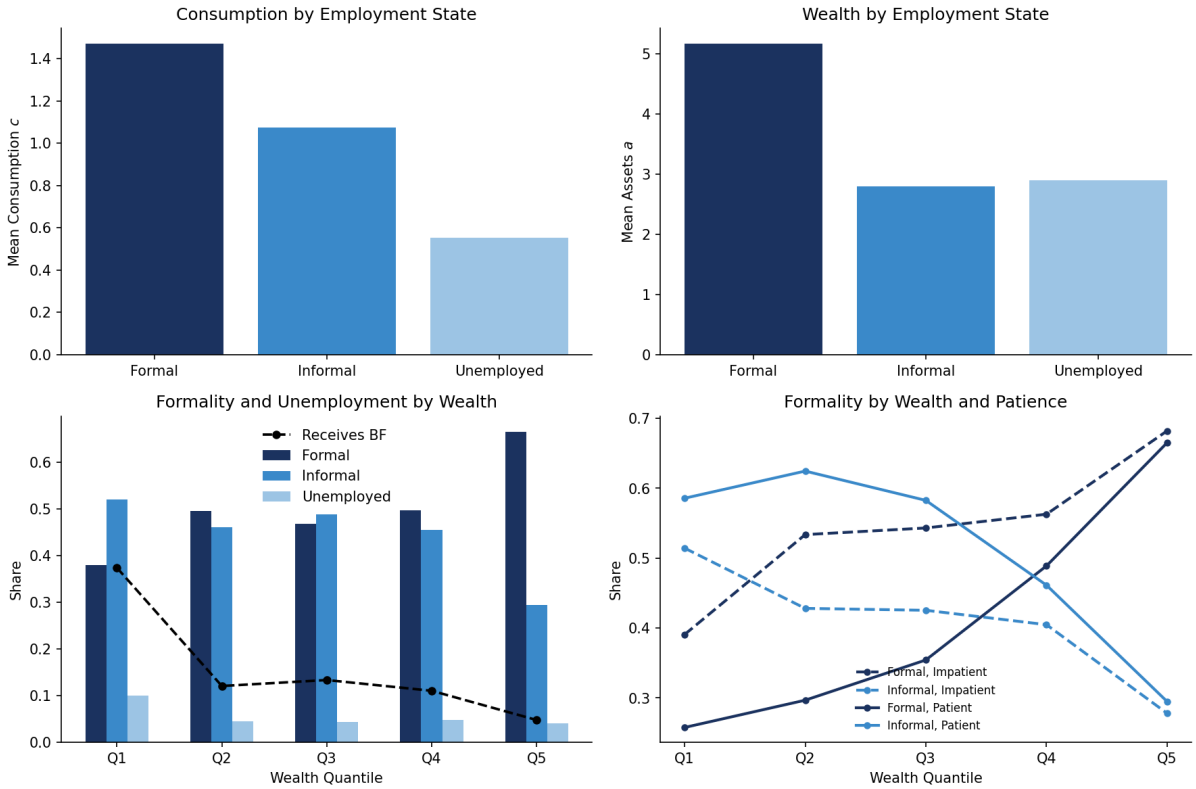


Figure 12: Steady-State Descriptive Statistics

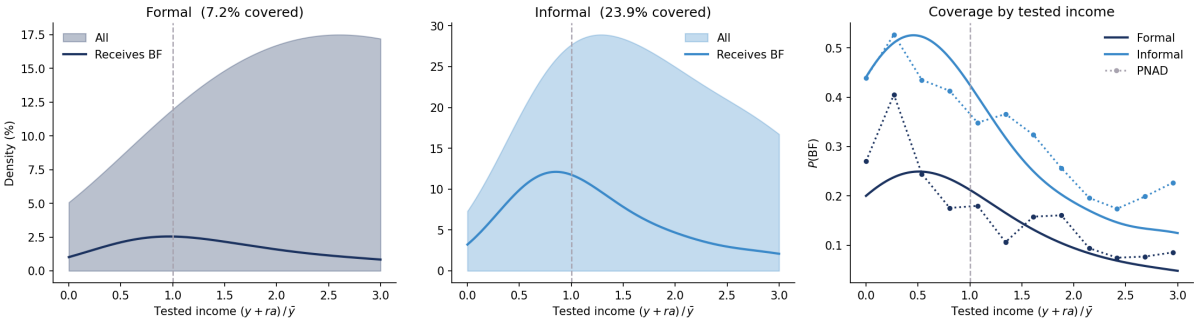


Figure 13: Steady-State Distribution of BF Receipts

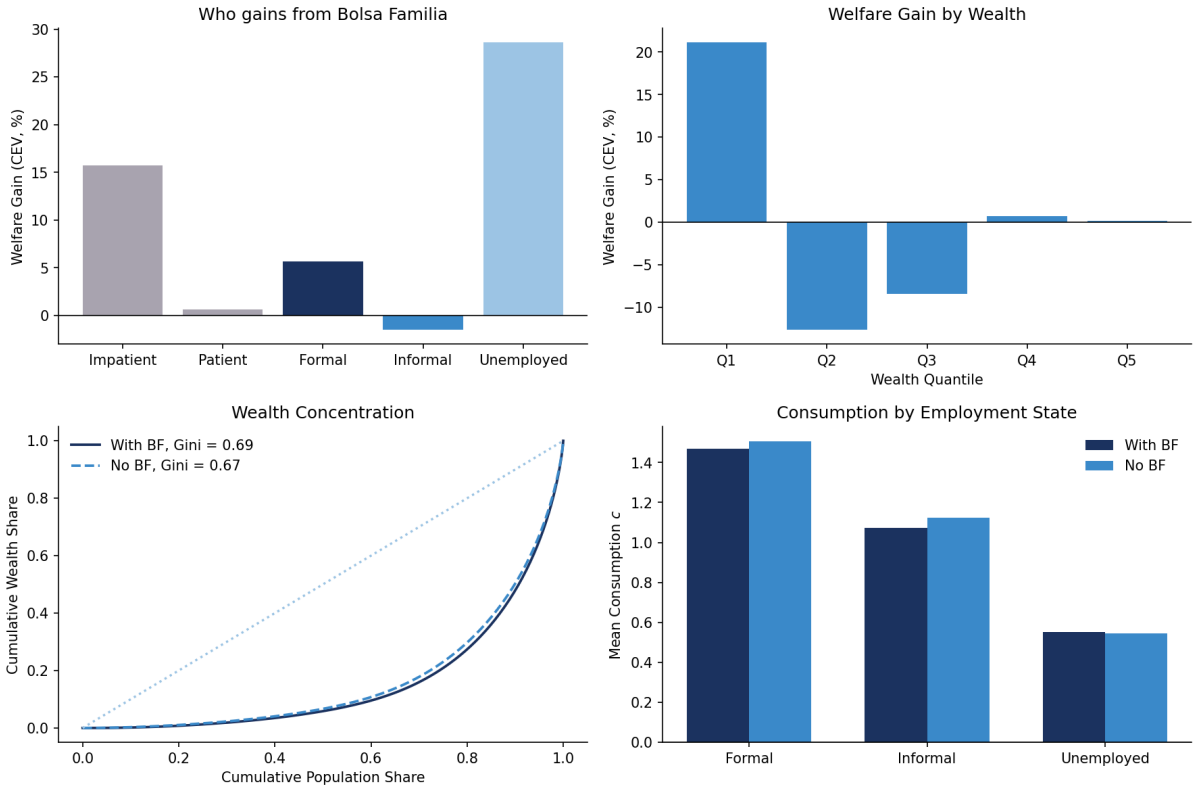


Figure 14: Steady-State Comparison: With and Without BF

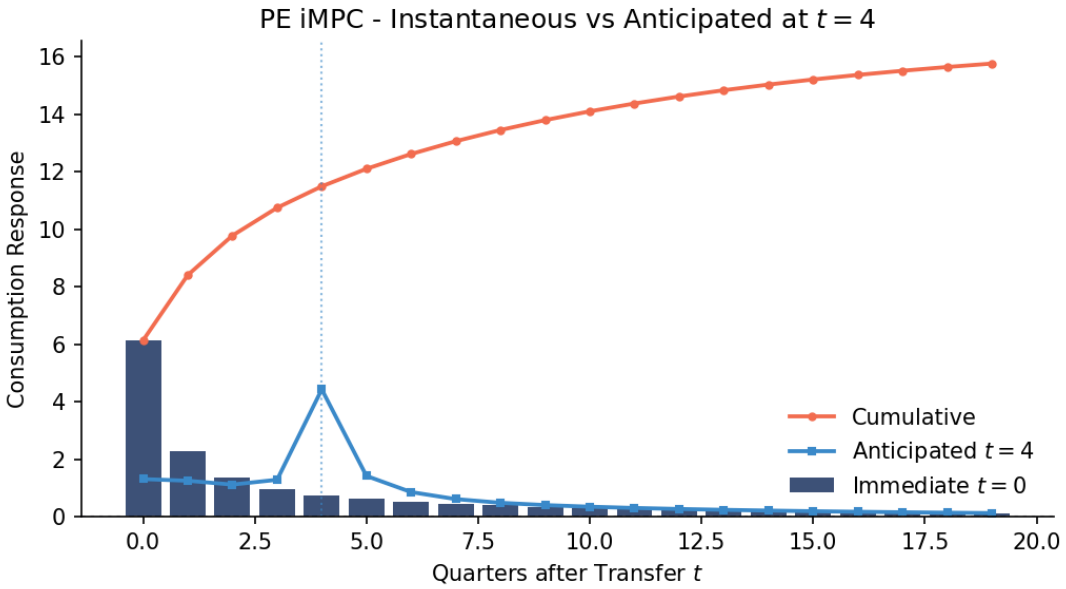


Figure 15: iMPC: Partial Equilibrium Profile

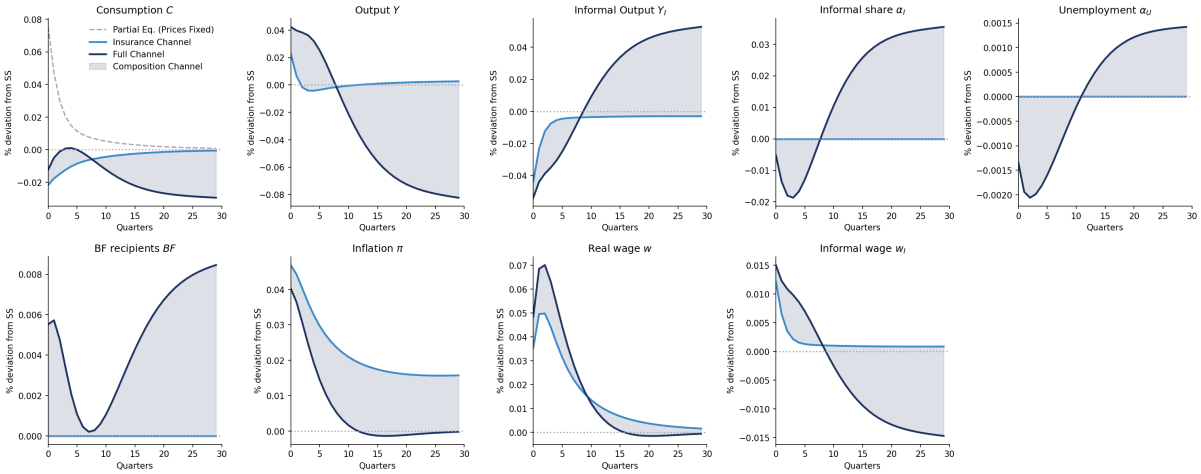


Figure 16: Response to a Shock in T by Channel

Table 8: Cumulative Response (%), Shock in T

Variable	Horizon	Insurance	Full	Composition
Consumption C	1	-0.022	-0.012	0.009
	4	-0.067	-0.018	0.049
	20	-0.138	-0.245	-0.107
	100	0.329	-3.863	-4.192
Unemployment α_U	1	0.000	-0.001	-0.001
	4	-0.000	-0.007	-0.007
	20	-0.000	-0.008	-0.008
	100	-0.000	0.214	0.214
Inflation π	1	0.047	0.040	-0.007
	4	0.168	0.132	-0.036
	20	0.508	0.185	-0.323
	100	3.365	2.221	-1.143
Real wage w	1	0.035	0.048	0.013
	4	0.179	0.250	0.071
	20	0.408	0.479	0.071
	100	0.961	0.684	-0.277

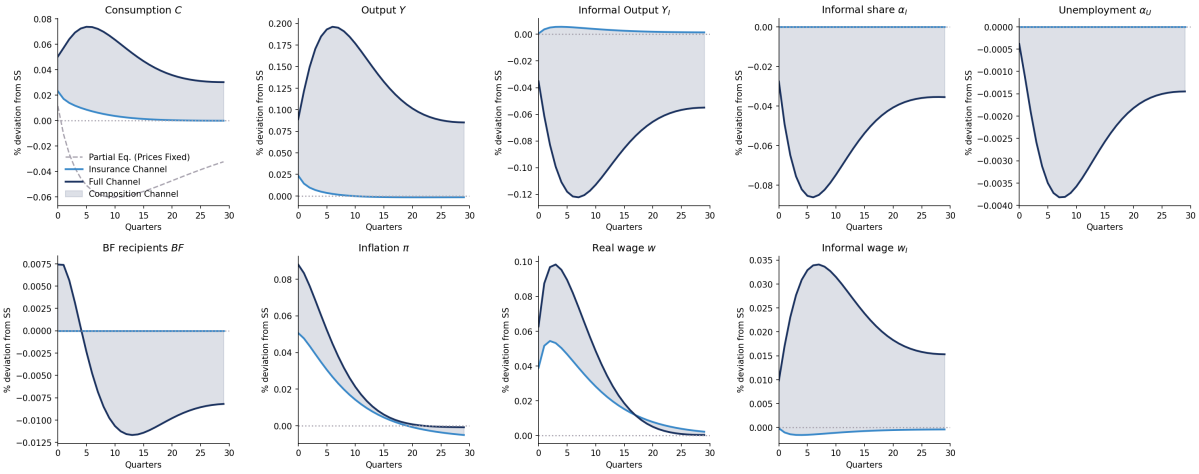


Figure 17: Response to an reduction in i by Channel

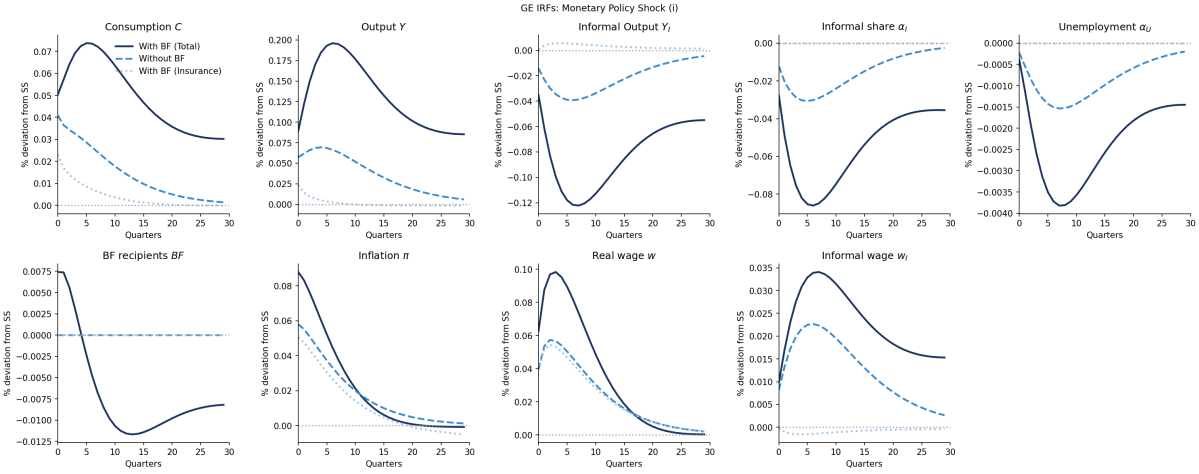


Figure 18: Response to an reduction in i: With and Without CCT

Robustness Tests

Lastly, in order to verify whether this results are representatives and robust, we intent to do some robustness tests changing some calibration parameters, such as γ, φ, r^* (4% and 6%), ϕ (2.0), the informal returns to scale α_I, and the informal wage flexibility κ_w^I. These parameters might have minor implications in the results, but it is expected that the model continues to lead to the same results.

Another test would be removing self-employed and unpaid family (Conta Própria and Familiar Auxiliar) from informal households in PNAD.

Limitations and Extensions

Formality in Brazil also buys Unemployment Insurance and FGTS, which are extra insurance advantages for formal workers. This model does not add this feature, so it

Table 9: Cumulative Response (%), Shock in i

Variable	Horizon	Insurance	Full	Composition
Consumption C	1	0.023	0.050	0.027
	4	0.066	0.241	0.175
	20	0.125	1.161	1.036
	100	-0.150	5.131	5.280
Unemployment α_U	1	0.000	-0.000	-0.000
	4	-0.000	-0.006	-0.006
	20	-0.000	-0.055	-0.055
	100	-0.000	-0.343	-0.343
Inflation π	1	0.051	0.088	0.037
	4	0.181	0.316	0.135
	20	0.394	0.657	0.264
	100	-0.907	-2.851	-1.944
Real wage w	1	0.039	0.063	0.024
	4	0.198	0.345	0.147
	20	0.616	1.044	0.428
	100	0.428	0.481	0.054

understates the value of formality and therefore overstates the composition channel. This is a natural extension of this paper.

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Appendix

Preference Shocks

As in Caliendo, Dvorkin, and Parro (2019), before deciding, the households receive an offer $A \in \{F, I, FI\}$ and draws a vector $\{\varepsilon^j\}$ over the options $j \in \mathcal{S}$, with scale σ_S[7], and picks the option j maximizing $V^j + \varepsilon^j$. This delivers a closed-form expected value,

$$\mathbb{P}^A(j) = \frac{\exp(V^j/\sigma_S)}{\sum_{k\in\mathcal{S}} \exp(V^k/\sigma_S)}, \qquad \mathbb{E}\Big[\max_{j\in\mathcal{S}} \left\{V^j + \varepsilon^j\right\}\Big] = \sum_{j\in\mathcal{S}} \mathbb{P}^A(j) V^j.^{8}$$

These shocks render the transition a differentiable function, which allows it to be linearized when computing the sequence-space Jacobians, as in Iskhakov et al. (2017). Therefore, the sector transition matrix $\Pi_t^s = \mathbb{P}(s'|s)$ is given by

$$\Pi_t^s = \begin{bmatrix} (1-\delta_F)\phi_F(F) & (1-\delta_F)\phi_F(I) & \delta_F \\ (1-\delta_I)\phi_I(F) & (1-\delta_I)\phi_I(I) & \delta_I \\ \phi_U(F) & \phi_U(I) & \phi_U(U) \end{bmatrix},$$

where $\phi_s(s')$ is the probability of accepting an offer s' given that the worker is in sector s,

$$\begin{aligned}
\phi_F(I) &= \pi_F \pi_I \, \mathbb{P}^{FI}(I) + (1-\pi_F)\pi_I \, \mathbb{P}^I(I), & \phi_F(F) &= 1 - \phi_F(I), \\
\phi_I(F) &= \pi_F \pi_I \, \mathbb{P}^{FI}(F) + \pi_F(1-\pi_I) \, \mathbb{P}^F(F), & \phi_I(I) &= 1 - \phi_I(F), \\
\phi_U(F) &= \pi_F^U \pi_I^U \, \mathbb{P}^{FI}(F) + \pi_F^U(1-\pi_I^U) \, \mathbb{P}^F(F), & & \\
\phi_U(I) &= \pi_F^U \pi_I^U \, \mathbb{P}^{FI}(I) + (1-\pi_F^U)\pi_I^U \, \mathbb{P}^I(I), & \phi_U(U) &= 1 - \phi_U(F) - \phi_U(I).
\end{aligned}$$

The preference shock smoothing parameter is set to $\sigma_S = 0.600$, the least possible value such that the steady state converged.

Price Phillips Curve

Let Y_t be the aggregate output of the economy, a CES of the intermediary inputs,

$$Y_t = \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} \, dj \right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

This firm maximizes its output subject to $\int_0^1 p_{i,j} y_{i,j} \, dj$. The FOC implies that the demand for the intermediate good is given by

$$y_{j,t} = \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} Y_t,$$

where P_t is the equilibrium price of the final good,

$$P_t = \left(\int_0^1 p_{j,t}^{1-\varepsilon} \, dj \right)^{\frac{1}{1-\varepsilon}}.$$

[7] σ_S governs the smoothness of the decision. As $\sigma_S \to 0$ it collapses to the deterministic max.

[8] σ_S is scaled by the marginal value of wealth, V_a, so that the randomness of the sectoral choice does not vary mechanically with wealth.

We assume a Rotemberg (1982) quadratic price adjustment cost given by

$$\frac{\theta_p}{2}\left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi}\right)^2 Y_t,$$

where $\theta_p = (\varepsilon - 1)/\kappa$ is the adjustment cost multiplier and $\bar{\Pi}$ the steady-state inflation.

Let $\Xi(y)$ be the intermediary firm's cost. Given last period's price $p_{j,t-1}$, the firm chooses this period's price $p_{j,t}$ to maximize the present discounted values of future profits,

$$V_t^{IF}(p_{j,t-1}) = \max_{p_{j,t}} \frac{p_{j,t}}{P_t} y_{j,t} - \Xi(y_{j,t}) - \frac{\theta_p}{2}\left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi}\right)^2 Y_t + \frac{1}{1+r_{t+1}} V_{t+1}^{IF}(p_{j,t}).$$

We can replace the demand function into this problem, so

$$V_t(p_{j,t-1}) = \max_{p_{j,t}} \frac{p_{j,t}}{P_t}\left(\frac{p_{j,t}}{P_t}\right)^{-\varepsilon} Y_t - \Xi(y_{j,t}) - \frac{\theta_p}{2}\left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi}\right)^2 Y_t + \frac{1}{1+r_{t+1}} V_{t+1}(p_{j,t}).$$

The FOC with respect to $p_{j,t}$ is

$$(1-\varepsilon)\left(\frac{p_{j,t}}{P_t}\right)^{-\varepsilon}\frac{Y_t}{P_t} + \varepsilon mc_{j,t} - \theta_p\left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi}\right)\frac{Y_t}{p_{j,t}} + \frac{1}{1+r_{t+1}} V'_{t+1}(p_{j,t}) = 0,$$

where $mc_t = \partial\Xi/\partial y_{j,t}$, and the envelope condition gives us

$$V'_{t+1} = \theta_p\left(\frac{p_{j,t+1}}{p_{j,t}} - \bar{\Pi}\right)\frac{p_{j,t+1}}{p_{j,t}}\frac{Y_{t+1}}{p_{j,t}}.$$

Thus, taking $p_{j,t} = P_t$ in equilibrium,

$$\theta_p(\pi_t - \bar{\Pi})\pi_t = \varepsilon\left(mc_t - \frac{\varepsilon-1}{\varepsilon}\right) + \frac{1}{1+r_{t+1}}\theta_p(\pi_{t+1} - \bar{\Pi})\pi_{t+1}\frac{Y_{t+1}}{Y_t},$$

where we define the firm's markup as $\mu = \frac{\varepsilon}{\varepsilon-1}$. Linearizing around the steady-state $\pi_t = \bar{\Pi}$,

$$\ln(1+\pi_t) = \frac{\varepsilon}{\theta_p}\left(mc_t - \frac{1}{\mu}\right) + \frac{1}{1+r_{t+1}}\ln(1+\pi_{t+1})\frac{Y_{t+1}}{Y_t},$$

where $\kappa = \varepsilon/\theta_p$ and $mc_t = w_t/Z_t$. This environment is equivalent to Calvo (1983) in first order around zero inflation.

Wage Phillips Curve

We will use the same setting as Hagedorn, Manovskii, and Mitman (2019), but considering just the formal sector. Assume there is a union that aggregate effective labor input into

$$h_t = \left(\int_0^1 e_{i,t}\theta_{i,t}(h_{i,t})^{\frac{\varepsilon_w - 1}{\varepsilon_w}}\, di\right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}},$$

sells households labor services $e_{i,t}\theta_{i,t}h_{i,t}$ at nominal wage $W_{i,t}$ to the labor recruiter, which minimizes costs $\int_0^1 W_{i,t}e_{i,t}\theta_{i,t}h_{i,t}\,di$, implying a demand of

$$h_{i,t} = \left(\frac{W_{i,t}}{W_t}\right)^{-\varepsilon_w} h_t.$$

The unions sets a nominal wage $\hat{W}_t$ for an effect unit of labor, $W_{i,t} = \hat{W}_t$ to maximize profits subject to Rotemberg (1982)'s wage adjustment costs

$$e_{i,t}\theta_{i,t}\frac{\theta_w}{2}\left(\frac{\hat{W}_t}{\hat{W}_{t-1}} - \bar{\Pi}^w\right)^2 h_t.$$

We assume that the union's wage setting problem is to maximize the aggregate utility of its members, so

$$V_t^w = \max_{\hat{W}_t} \int \left[u'(c_{i,t})\frac{(1-\tau_\ell)\hat{W}_t}{P_t}e_{i,t}\theta_{i,t}h_{i,t} - v(h_{i,t}) - e_{i,t}\theta_{i,t}h_t\frac{\theta_w}{2}\left(\frac{\hat{W}_t}{\hat{W}_{t-1}} - \bar{\Pi}^w\right)^2\right] d\Lambda_t + \bar{\beta}_F V_{t+1}^w.$$

Using the same algebra as the price Phillips curve, assuming that the union allocates the same amount of hours in each worker $h_{i,t} = h_t$, and using $\int_F d\Lambda_t = F_t$, the size of the formal sector,

$$\theta_w(\pi_t^w - \bar{\Pi}^w)\pi_t^w = \varepsilon_w\left[v'(h_t) - \frac{\varepsilon_w - 1}{\varepsilon_w}(1-\tau_\ell)w_t\Lambda_t^F\right] + \bar{\beta}\theta_w(\pi_{t+1}^w - \bar{\Pi}^w)\pi_{t+1}^w\frac{h_{t+1}}{h_t},$$

where $\Lambda_t^F = \int_F e_{i,t}\theta_{i,t}u'(c_{i,t})d\Lambda_t/F_t$. We can linearize around the steady state $\pi_t^w = \bar{\Pi}^w$ and take $h_{t+1} = h_t$, since there is no population nor technological growth. Thus,

$$\ln(1+\pi_t^w) = \frac{\varepsilon_w}{\theta_w}\left[v'(h_t) - (1-\tau_\ell)w_t\Lambda_t^F\frac{\varepsilon_w - 1}{\varepsilon_w}\right] + \bar{\beta}\ln(1+\pi_{t+1}^w).$$

Therefore, defining $\kappa_w = (\varepsilon_w/\theta_w)$ at the steady state and $\mu_w = \varepsilon_w/(\varepsilon_w - 1)$, the wage Phillips curve is

$$\ln(1+\pi_t^w) = \kappa_w\left[\psi \cdot h_t^{1/\varphi} - \frac{(1-\tau_\ell)w_t}{\mu_w F_t}\int_F e_{i,t}\theta_{i,t}u'(c_{i,t})d\Lambda_t\right] + \bar{\beta}\ln(1+\pi_{t+1}^w).$$

At the same time, the informal union's problem is exactly the same, but without a wage tax and with a different marginal revenue given by $\alpha_I w_t^I$. Assuming a different adjustment cost θ_w^I, the wage Phillips curve for informal wage is

$$\ln(1+\pi_t^{w_I}) = \kappa_w^I\left[\psi \cdot \left(h_t^I\right)^{1/\varphi} - \frac{\alpha_I w_t^I}{\mu_w I_t}\int_I e_{i,t}\theta_{i,t}u'(c_{i,t})d\Lambda_t\right] + \bar{\beta}_I\ln(1+\pi_{t+1}^{w_I}).$$

Computational Appendix

Table 10: SMM Estimates and Over-identification Test

Parameter	Estimate	S.E.	t	p	Lower	Upper
mu_I	0.036	0.094	0.38	0.705	-1.00	0.300
sigma_F	0.422	0.118	3.59	0.000	0.05	1.000
sigma_I	0.303	0.109	2.78	0.005	0.05	1.200
rho_e	0.956	0.027	35.07	0.000	0.70	0.999
sd_e	0.629	0.064	9.83	0.000	0.10	1.200
dbeta	0.293	0.098	2.98	0.003	0.00	0.400
Hansen J	6.29 (df 3, $p = 0.098$)					